Miscellaneous problems

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ST. XAVIER'S COLLEGIATE SCHOOL, 30 MOTHER TERESA SARANI, KOLKATA-16

Class:XI | **FIRST SEMESTER BLOCK TEST** | **Date: 02/09/2024**

Duration: Three hours | **COMPUTER SCIENCE PRACTICALS** | **Full Marks: 30**

The total time to be spent on the Planning Session and the Examination Session is **Three hours**. After completing the Planning Session, the candidate may begin with the Examination Session. A maximum of 90 minutes is permitted for the Planning Session. However, if candidates finish earlier, they are to be permitted to begin the Examination Session.

This paper consists of three problems from which candidates are required to attempt any **one** problem. Candidates are expected to do the following:

A. Planning Session:

1. Write an algorithm for the selected problem. [3 marks]
(Algorithm should be expressed clearly using any standard scheme such as pseudo code or in steps which are simple enough to be obviously computable.)
2. Write a program in JAVA language. The program should follow the algorithm and should be logically and syntactically correct. Document the program using mnemonic names / comments, identifying and clearly describing the choice of data types and meaning of variables. [7 marks]

B. Examination Session:

1. Code / Type the program on the computer. Typically, this should be a program that compiles and runs correctly. [2 marks]
2. Test run the program on the computer using the given sample data and get the output in the format specified in the problem. [3 marks]

Note: The candidates must not carry any stationery, items such as pen / pencil / eraser to the Computer Laboratory for the Examination Session.

Solve any **one** of the following Problems.

Question 1:

Given two positive numbers M and N, such that M is between 100 and 10000 and N is less than 100. Find the smallest integer that is greater than M and whose digits add up to N. For example, if M = 100 and N = 11, then the smallest integer greater than 100 whose digits add up to 11 is 119.

Write a program to accept the numbers M and N from the user and print the smallest required number whose sum of all its digits is equal to N. Also, print the total number of digits present in the required number. The program should check for the validity of the inputs and display an appropriate message for an invalid input.

Test your program with the sample data and some random data:

Example 1
INPUT :
M = 100
N = 11
OUTPUT :
The required number = 119
Total number of digits = 3

Example 2
INPUT :
M = 1500
N = 25

OUTPUT :
The required number = 1699
Total number of digits = 4

OUTPUT :
The required number = 1699
Total number of digits = 4

Example 3
INPUT :
M = 99
N = 11

OUTPUT :
INVALID INPUT

Example 4
INPUT :
M = 112
N = 130

OUTPUT :
INVALID INPUT

Question 2

Write a program to declare a matrix A[ ][ ] of order (M x N) where 'M' is the number of rows and 'N' is the number of columns such that the value of 'M' must be greater than 0 and less than 10 and the value of 'N' must be greater than 2 and less than 6. Allow the user to input digits (0 - 7) only at each location, such that each row represents an octal number.

Example: 2 3 1 (decimal equivalent of 1st row = 153 i.e. $2x8^2 + 3x8^1 + 1x8^0$)
4 0 5 (decimal equivalent of 2nd row = 261 i.e. $4x8^2 + 0x8^1 + 5x8^0$)
1 5 6 (decimal equivalent of 3rd row = 110 i.e. $1x8^2 + 5x8^1 + 6x8^0$)

Perform the following tasks on the matrix:
(a) Display the original matrix.
(b) Calculate the decimal equivalent for each row and display as per the format given below.
Test your program for the following data and some random data:

Example 1
INPUT: M = 1
N = 3
ENTER ELEMENTS FOR ROW 1: 1 4 4

OUTPUT:	FILLED MATRIX	DECIMAL EQUIVALENT
	1 4 4	100

Example 2
INPUT: M = 3
N = 4
ENTER ELEMENTS FOR ROW 1: 1 1 3 7
ENTER ELEMENTS FOR ROW 2: 2 1 0 6
ENTER ELEMENTS FOR ROW 3: 0 2 4 5

OUTPUT:	FILLED MATRIX	DECIMAL EQUIVALENT
	1 1 3 7	607
	2 1 0 6	1094
	0 2 4 5	165

Question 3

Write a program to accept a date in the string format dd/mm/yyyy and accept the name of the day on 1st of January of the corresponding year. Find the day for the given date.

Example:

INPUT

of the corresponding year. Find the day for the given date.

Example:

INPUT

Date : 5/7/2024 Day on 1st January : MONDAY

OUTPUT

Day on 5/7/2024 : FRIDAY

Test run the program on the following inputs.

INPUT DATE	DAY ON 1ST JANUARY	OUTPUT DAY FOR INPUT DATE
4/9/1998	THURSDAY	FRIDAY
31/8/1999	FRIDAY	TUESDAY
6/12/2000	SATURDAY	WEDNESDAY

The program should include the part for validating the inputs namely the date.

LA MARTINIERE FOR BOYS

CLASS – 12

SELECTION EXAMINATION

COMPUTER SCIENCE

Paper – 2

(PRACTICAL)

(Maximum Marks: 70)

(Time allowed: Three hours)

(Candidates are allowed additional 15 minutes for only reading the paper.

They must NOT start writing during this time.)

The total time to be spent on the Planning Session and the Examination Session is three hours.

Planning session: 90 minutes

Examination session: 90 minutes

Note: Candidates are to be permitted to proceed to the Examination Session only after 90 minutes of the Planning session are over.

*This paper consists of **three** problems from which candidates are required to attempt **any one** problem.*

Candidates are expected to do the following:

1. Write a program in Java.
2. Write an Algorithm.
3. Document the Program with mnemonic names/ comments
4. Write Variable Listing
5. Code/Type the program in the computer and take a print out.

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Question 1

Write a program to accept a number (less than or equal to 12 digits) and display the longest sequence of consecutive natural numbers in the forward direction or in the backward direction . Also print the sum of the sequence.

Sample Input : 9156782678
Sample Output:
Longest sequence : 5678
Sum of the sequence 26

Sample Input : 987675429
Sample Output :
Largest sequence 6789
Sum of the sequence 30

Question 2

Write a program to accept elements in to a 2 dimensional matrix of m rows and n columns (m<=10 , n<=12) and rotate the matrix in a clockwise manner row-wise. The first row becomes the last row and the last row becomes the second last row and it continues in the manner, so that the second row becomes the first row

Consider an example :
m=3 n=3
Elements in the matrix

1	2	3
4	5	6
7	8	9

After 1 rotation the matrix will look like

4	5	6
7	8	9
1	2	3

The maximum number of permissible rotations is m-1

Sample Input : m=4 n=5
Number of rotations : 2

Sample Input : m=4 n=5
Number of rotations : 2
Enter elements

1	2	3	4	5
6	7	8	9	1
2	3	4	5	6
7	8	9	1	2

Sample Output:

2	3	4	5	6
7	8	9	1	2
1	2	3	4	5
6	7	8	9	1

Sample Input :m=3 n=5

Number of rotations : 5

Sample Output : Invalid

Question 3

Write a program to accept a Roman number between [1-2000]in uppercase and print in Indo-Arabic numerals.

Note: I=1 V=5 X=10 L=50 C=100 D=500 M=1000

Sample

Input : MCMXCVIII

Output : 1998

Input : CMXCIX

Output : 999